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Robust Multi-Asset VaR and Expected Shortfall Forecasting

via GJR-GARCH, Extreme Value Theory, and Vine Copula Model:
Evidence from Indian Equity Markets

A Dual Degree Project

Submitted by

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THESIS CERTIFICATE

This is to certify that the Dual Degree Project titled **ROBUST MULTI-ASSET VAR AND EXPECTED SHORTFALL FORECASTING VIA GJR-GARCH, EXTREME VALUE THEORY, AND VINE COPULA MODELS: EVIDENCE FROM INDIAN EQUITY MARKETS**, submitted by **N. Devendra (ME21B127)** to the Indian Institute of Technology Madras, for the award of the Interdisciplinary Dual Degree in Quantitative Finance, is a bona fide record of the research work done under the supervision of **Prof. Neelesh Upadhye**. The contents of this Dual Degree Project, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Abstract

This report develops and empirically evaluates a fallback-aware framework for forecasting portfolio Value-at-Risk (VaR) and Expected Shortfall (ES) in a rolling multi-asset setting, with application to Indian equity markets. The central challenge addressed is the practical problem of Extreme Value Theory (EVT) threshold selection instability: EVT-based tail modelling relies on the selection of sufficiently high and statistically reliable thresholds for Generalized Pareto Distribution (GPD) fitting, yet such thresholds may not exist in all rolling windows. Consequently, fallback mechanisms become operationally necessary for maintaining continuity in rolling risk forecasting systems.

The framework combines EVT-based tail modelling with clearly defined fallback approaches—a KDE-based method and a model-implied (no-KDE) alternative—so that the model remains usable even when EVT fails. Dependence across assets is captured using vine copulas, and the entire pipeline is applied in a rolling-window setup across different portfolio sizes (two and four assets) and forecast horizons (one day and ten days).

Model performance is evaluated using the mean Fissler–Ziegel (FZ0) joint VaR–ES loss as the primary criterion, supplemented by standard backtests including Kupiec, Christoffersen, Dynamic Quantile, and Acerbi–Székely. Results show that no single model dominates across all configurations; performance is context-dependent, varying with forecast horizon and portfolio size. Differences in FZ0 values are small (maximum approximately 0.037).

These findings suggest that more complex tail models do not uniformly outperform simpler baseline models, which remain competitive across several configurations. The results also establish that fallback mechanisms are operationally necessary in rolling EVT-based systems, and that the design of the fallback policy—not just the tail model itself—can influence forecast quality.

KEYWORDS: Value-at-Risk; Expected Shortfall; Extreme Value Theory; Fallback Modeling; Vine Copula; Fissler–Ziegel Loss; Indian Equity Markets

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Chapter 1

Introduction

Accurate measurement of market risk remains a central challenge in quantitative finance. Value-at-Risk (VaR), which measures the maximum expected loss at a given confidence level over a specified horizon, and Expected Shortfall (ES), which measures the expected loss conditional on losses exceeding the VaR threshold, are the primary regulatory risk measures under the Basel III and IV frameworks. However, their practical estimation for multi-asset equity portfolios is complicated by features such as volatility clustering, heavy tails, and asymmetric dependence. These characteristics are not adequately captured by standard normal-distribution assumptions in traditional models, leading to potential underestimation of extreme losses.

To address these limitations, a substantial body of literature combines GARCH-class volatility models with Extreme Value Theory (EVT) for tail estimation and copula models for capturing multivariate dependence. However, many applied studies focus on developed markets, and EVT threshold selection is often handled using fixed or heuristic rules rather than being treated as a dynamic rolling-window problem.

1.1 Motivation and Context

In practice, EVT-based modelling presents an additional challenge: the threshold required for fitting the Generalised Pareto Distribution (GPD) must be re-selected in each rolling window, and its reliability may vary over time. In some windows, no threshold may satisfy the required goodness-of-fit, minimum exceedance, and parameter stability criteria for reliable GPD estimation. This raises a practical question: how should tail risk be modelled in windows where EVT threshold selection becomes unreliable? Despite its importance for real-world risk systems, this issue remains relatively underexplored in the rolling-window risk literature. This report addresses these challenges by developing and empirically evaluating a rolling multi-asset risk forecasting framework that combines GPD-based EVT tail modelling, fallback-aware estimation, and vine copula dependence modelling. The framework is applied to portfolios constructed from stocks listed on the National Stock Exchange of India (NSE), specifically SBI, Infosys, HUL, and Tata Motors.

1.2 Objectives

The key objectives of this report are as follows:

1. To explicitly address EVT threshold failure in a rolling-window setting, identifying windows where threshold selection fails and handling them directly through a fallback-aware framework.
2. To define and compare two fallback approaches—a KDE-based method and a model-implied (no-KDE) alternative—within a multi-asset VaR/ES framework, and to study their impact on risk forecasts.
3. To adopt an overlap-aware evaluation framework for multi-day forecasts, ensuring valid statistical inference when independence-based backtests cannot be applied.
4. To provide empirical evidence on Indian equity portfolios across multiple configurations, using the mean Fissler–Ziegel (FZ0) joint loss as the primary evaluation metric.

The remainder of this report is organised as follows. Chapter 2 reviews the relevant literature. Chapter 3 describes the data and provides tail behaviour diagnostics. Chapter 4 presents the full methodology, covering the return model, EVT threshold selection, fallback regimes, hybrid CDF construction, vine copula fitting, and the evaluation framework. Chapter 5 reports the empirical results. Chapter 6 discusses the findings. Chapter 7 outlines managerial implications. Chapter 8 concludes and outlines directions for future work.

Chapter 2

Literature Review

The motivation for advanced Value-at-Risk (VaR) modelling arises from the limitations of normality-based approaches. This chapter surveys the foundational and recent literature across three interconnected streams: volatility modelling, extreme value theory, and copula-based dependence modelling.

2.1 Volatility Modelling

Early studies by Mandelbrot (1963) and Fama (1965) showed that financial return distributions are heavy-tailed, leading to underestimation of extreme losses under standard models. To account for time-varying volatility, Engle (1982) introduced the ARCH model, extended by Bollerslev (1986) to the GARCH framework. Glosten et al. (1993) further extended this through the GJR-GARCH model, which captures the asymmetric impact of negative versus positive shocks on conditional variance—a feature commonly observed in equity markets and directly relevant to risk estimation.

2.2 Extreme Value Theory in Risk Modelling

A common approach for modelling tail risk combines GARCH filtering with Extreme Value Theory (EVT) using the Peaks-over-Threshold (POT) method (McNeil and Frey, 2000; Embrechts et al., 1997). EVT provides a theoretically justified framework for modelling extremes, with the Generalised Pareto Distribution (GPD) established as the limiting distribution for exceedances above a high threshold (Balkema and de Haan, 1974; Pickands, 1975).

However, the practical performance of EVT depends heavily on threshold selection. Issues related to threshold instability in rolling settings are not always explicitly addressed in applied risk models; many studies rely on fixed quantile-based thresholds such as the 90th, 95th, or 99th percentile. While convenient, such thresholds may not remain statistically valid across rolling windows as tail behaviour changes over time. This report directly addresses this gap by incorporating automated multi-criterion threshold selection with explicit fallback handling.

2.3 Copula-Based Dependence Modelling

In multivariate settings, dependence across assets is modelled using copulas. Sklar’s theorem (Sklar, 1959) provides the foundation for separating marginal distributions from dependence structures. In practice, standardised residuals obtained after GARCH filtering and EVT-based marginal modelling are transformed using the Probability Integral Transform (PIT) and subsequently used for dependence modelling. Vine copula constructions (Bedford and Cooke, 2001; Czado, 2019) extend these ideas to higher dimensions and are commonly used in multi-asset portfolio risk modelling, offering flexibility to capture asymmetric and tail-dependent structures.

2.4 Model Evaluation

The evaluation of risk models has received considerable attention. Kupiec (1995) and Christoffersen (1998) developed likelihood-based tests for VaR exceedances and their independence. Engle and Manganelli (2004) proposed the Dynamic Quantile (DQ) test for detecting dynamic misspecification. For Expected Shortfall, Acerbi and Székely (2014) introduced simulation-based testing methods. More recently, Fissler and Ziegel (2016) showed that VaR and ES are jointly elicitable, enabling the use of consistent scoring functions such as the FZ0 loss, which serves as the primary evaluation criterion in this report.

In multi-day forecasting settings, overlapping observations introduce dependence, which requires careful interpretation of independence-based backtests—a consideration explicitly incorporated into the evaluation framework of this study.

Chapter 3

Data

3.1 Assets and Sample Period

The study uses daily closing prices sourced from Yahoo Finance for four assets listed on the National Stock Exchange of India: State Bank of India (SBIN.NS), Infosys Limited (INFY.NS), Hindustan Unilever Limited (HINDUNILVR.NS), and Tata Motors Limited (TATAMOTORS.NS). These assets represent distinct economic sectors—banking, information technology, consumer staples, and automotive manufacturing—providing meaningful diversification in the four-asset portfolio setting. The two-asset configuration uses SBIN.NS and INFY.NS.

The full data span covers April 27, 2011 to May 27, 2025, yielding approximately 3,472 trading days. Log returns are computed as $r_t = 100 \times \ln(P_t/P_{t-1})$, where P_t denotes the closing price at time t , and are reported as percentage log-returns. The first 1,000 observations form the initial estimation window, after which a rolling-window approach is employed with the model re-estimated in each window.

3.2 Tail Behaviour Diagnostics

Before presenting the estimation framework, we examine the tail behaviour of the standardised residuals to assess whether a fitted parametric model adequately captures extreme risk. Tail diagnostics are based on Q–Q plots and Hill estimates computed from an AR(1)-GJR-GARCH(1,1) fit with Student- t innovations over the initial estimation window.

The Hill estimator (Hill, 1975) provides a measure of tail heaviness through the extreme value index $\hat{\gamma}$, defined as:

$$\hat{\gamma}_k = \frac{1}{k} \sum_{i=1}^k \left[\log X_{(n-i+1)} - \log X_{(n-k)} \right] \quad (3.1)$$

where k denotes the number of upper order statistics (extreme observations) used in the estimation, with $1 \leq k < n$; n is the sample size; $X_{(i)}$ represents the i -th order statistic of the sample sorted in ascending order, i.e., $X_{(1)} \leq \dots \leq X_{(n)}$; $X_{(n-i+1)}$ is the i -th largest observation; $X_{(n-k)}$ is the $(k+1)$ -th largest observation and acts as a data-dependent threshold. The estimator computes the average log-excess of the largest k observations over the threshold $X_{(n-k)}$, providing a measure of tail heaviness.

Table 3.1 summarises the Hill estimates for both upper and lower tails across all assets. The estimated values range approximately from 0.32 to 0.47, indicating substantial tail heaviness. In comparison, the fitted Student- t distribution implies a tail index of $\gamma = 1/\nu$, lying between approximately 0.10 and 0.30 depending on the asset. The empirical Hill estimates are consistently higher than model-implied values, suggesting heavier tails than those captured by the fitted GJR-GARCH model.

Table 3.1: Hill Tail Index Estimates vs. Student- t (Initial Estimation Window)

Asset	Hill γ (Upper)	Hill γ (Lower)	$1/\nu$	Upper vs. t	Lower vs. t
HINDUNILVR.NS	0.3583	0.3873	0.2027	Heavier	Heavier
INFY.NS	0.4127	0.4730	0.3032	Heavier	Heavier
SBIN.NS	0.3650	0.3611	0.1488	Heavier	Heavier
TATAMOTORS.NS	0.3217	0.3258	0.1024	Heavier	Heavier

These findings are supported by Q-Q plot diagnostics, which show noticeable deviations in extreme quantiles from the fitted Student- t distribution. It is important to note that these diagnostics are based on a single estimation window and do not imply the same tail behaviour persists throughout the sample. Tail properties may vary across rolling windows, motivating the use of adaptive tail modelling with rolling threshold selection.

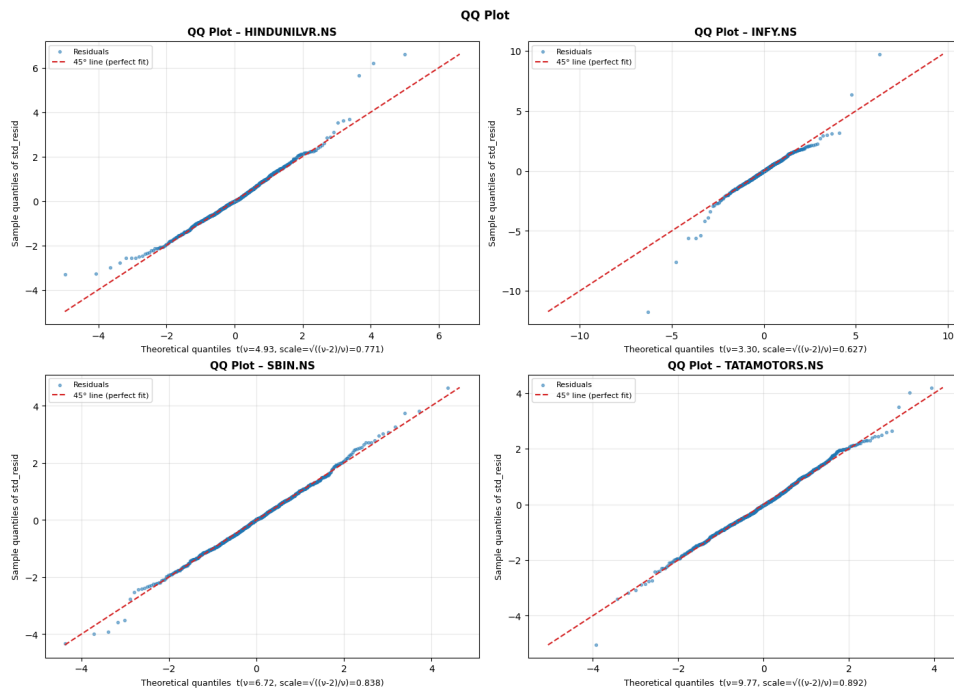


Figure 3.1: QQ Plot

Chapter 4

Methodology

For each rolling window, the following steps are executed in sequence: (i) AR(1)–GJR-GARCH(1,1) fitting, (ii) EVT threshold selection, (iii) fallback-aware hybrid CDF construction and Probability Integral Transform (PIT), (iv) vine copula fitting, and (v) Monte Carlo simulation for VaR and ES computation..

4.1 Return Model: AR(1)–GJR-GARCH(1,1)

For each asset i and each rolling window, returns r_t are modelled as:

$$r_t = \mu_t + \varepsilon_t, \quad \mu_t = c + \phi r_{t-1} \quad (4.1)$$

$$\varepsilon_t = \sigma_t z_t, \quad z_t \sim t(\nu) \quad (4.2)$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}_{\{\varepsilon_{t-1} < 0\}} + \beta \sigma_{t-1}^2 \quad (4.3)$$

where μ_t is the conditional mean, ε_t is the innovation, and σ_t^2 is the conditional variance. The innovations z_t follow a Student- t distribution with ν degrees of freedom. The leverage term $\gamma \varepsilon_{t-1}^2 \mathbf{1}_{\{\varepsilon_{t-1} < 0\}}$ captures the asymmetric impact of negative shocks on volatility. Parameters $\{c, \phi, \omega, \alpha, \gamma, \beta, \nu\}$ are estimated by maximum likelihood within each rolling window. Standardised residuals are computed as $\hat{z}_t = \varepsilon_t / \sigma_t$ and used for tail modelling.

As a benchmark, a Gaussian copula AR(1)–GJR-GARCH(1,1) model with the same mean and variance specification but normally distributed innovations and no EVT-based tail modelling is also considered.

4.2 EVT Threshold Selection

Threshold selection is a key step in Peaks-over-Threshold (POT) modelling. An automated multi-criterion procedure is implemented separately for the lower and upper tails of standardised residuals.

Threshold search: For each tail, 50 candidate thresholds are evaluated over the range corresponding to the 80th to 99.5th percentiles of the absolute residuals. For each candidate threshold u , exceedances above u are fitted using a GPD with parameters (ξ, σ) estimated by maximum likelihood.

Acceptance criteria: A threshold is accepted if it satisfies all of the following conditions:

- *Minimum exceedances:* at least 100 exceedances;
- *Kolmogorov–Smirnov test:* p -value > 0.05 ;
- *Anderson–Darling test:* bootstrap p -value > 0.05 (300 simulations);
- *Parameter stability:* estimated GPD parameters remain stable in a neighbourhood of the threshold.

Threshold selection outcome:

- **Full-pass case:** One or more thresholds satisfy all criteria; the threshold with the largest number of exceedances is selected and EVT is applied.
- **Sparse case:** A threshold satisfies goodness-of-fit and parameter stability but fails the minimum exceedance requirement; EVT is not considered reliable and fallback is triggered.
- **Invalid case:** No threshold satisfies both goodness-of-fit and parameter stability criteria; EVT is not applied and fallback is triggered.

4.3 Fallback Regimes

Fallback modelling is a central component of the framework. Based on the threshold selection procedure, each tail in each window is classified into full-pass, sparse, or invalid cases. Fallback is applied in sparse and invalid cases, while EVT is used only in full-pass cases.

Two alternative fallback approaches are evaluated:

- **No-KDE fallback (GJR-GARCH implied tails):** The tail is modelled using the parametric distribution implied by the AR(1)–GJR-GARCH(1,1) model. In sparse windows, the candidate threshold with the largest exceedance count is used; in invalid windows, the threshold with the maximum exceedances within the search range is selected. KDE is applied only to the central region.
- **KDE fallback:** When EVT thresholding is not reliable, KDE is applied to the entire distribution, including the affected tail, avoiding use of a poorly fitted GPD.

The with-KDE and no-KDE variants differ only in their fallback treatment; in full-pass windows, both approaches are identical.

4.4 Hybrid CDF and Probability Integral Transform

For each asset, a hybrid semi-parametric cumulative distribution function (CDF) is constructed by combining EVT-based GPD tails with a kernel density estimate (KDE) for the central region. The exact form of the hybrid CDF depends on whether EVT threshold selection succeeds in the lower and upper tails and on the fallback design adopted when EVT fitting is unreliable.

KDE Fallback Design

Case 1: EVT succeeds in both tails

$$F(x) = \frac{n_l}{n} (1 - G_L(u_L - x)), \quad x < u_L, \quad (4.4)$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n} \hat{F}_{\text{KDE}}(x), \quad u_L \leq x \leq u_U, \quad (4.5)$$

$$F(x) = 1 - \frac{n_u}{n} (1 - G_U(x - u_U)), \quad x > u_U. \quad (4.6)$$

Case 2: Lower-tail fallback, upper-tail EVT

$$F(x) = \left(1 - \frac{n_u}{n}\right) \hat{F}_{\text{KDE}}(x), \quad x \leq u_U, \quad (4.7)$$

$$F(x) = 1 - \frac{n_u}{n} (1 - G_U(x - u_U)), \quad x > u_U. \quad (4.8)$$

Case 3: Lower-tail EVT, upper-tail fallback

$$F(x) = \frac{n_l}{n} (1 - G_L(u_L - x)), \quad x < u_L, \quad (4.9)$$

$$F(x) = \frac{n_l}{n} + \left(1 - \frac{n_l}{n}\right) \hat{F}_{\text{KDE}}(x), \quad x \geq u_L. \quad (4.10)$$

Case 4: Fallback in both tails

$$F(x) = \hat{F}_{\text{KDE}}(x). \quad (4.11)$$

No-KDE Fallback Design

Case 1: EVT succeeds in both tails

Identical to the full-pass specification given previously.

Case 2: Lower-tail fallback, upper-tail EVT

$$F(x) = \frac{n_l}{n} F_{\text{GARCH}}(x), \quad x < u_L, \quad (4.12)$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n} \hat{F}_{\text{KDE}}(x), \quad u_L \leq x \leq u_U, \quad (4.13)$$

$$F(x) = 1 - \frac{n_u}{n} (1 - G_U(x - u_U)), \quad x > u_U. \quad (4.14)$$

Case 3: Lower-tail EVT, upper-tail fallback

$$F(x) = \frac{n_l}{n} (1 - G_L(u_L - x)), \quad x < u_L, \quad (4.15)$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n} \hat{F}_{\text{KDE}}(x), \quad u_L \leq x \leq u_U, \quad (4.16)$$

$$F(x) = 1 - \frac{n_u}{n} + \frac{n_u}{n} F_{\text{GARCH}}(x), \quad x > u_U. \quad (4.17)$$

Case 4: Fallback in both tails

$$F(x) = \frac{n_l}{n} F_{\text{GARCH}}(x), \quad x < u_L, \quad (4.18)$$

$$F(x) = \frac{n_l}{n} + \frac{n_c}{n} \hat{F}_{\text{KDE}}(x), \quad u_L \leq x \leq u_U, \quad (4.19)$$

$$F(x) = 1 - \frac{n_u}{n} + \frac{n_u}{n} F_{\text{GARCH}}(x), \quad x > u_U. \quad (4.20)$$

In the preceding expressions, u_L and u_U denote the lower and upper thresholds, respectively; n is the total number of observations; and n_l , n_c , and n_u are the numbers of observations assigned to the lower tail, central region, and upper tail, respectively, with $n = n_l + n_c + n_u$. The functions $G_L(\cdot)$ and $G_U(\cdot)$ denote the fitted Generalized Pareto Distribution (GPD) cumulative distribution functions for the lower and upper tails, applied to the exceedances $u_L - x$ and $x - u_U$, respectively. $\hat{F}_{\text{KDE}}(x)$ denotes the kernel density estimate (KDE)-based cumulative distribution function, appropriately normalized over the region over which KDE is applied. Finally, $F_{\text{GARCH}}(x)$ denotes the normalised parametric cumulative distribution function implied by the fitted AR(1)–GJR-GARCH(1,1) model with Student- t innovations.

Probability Integral Transform (PIT) values are computed for each asset from $F(x)$, transforming standardized residuals to uniform marginals on $[0, 1]$. Correct specification of the hybrid CDF implies that the PIT values should be uniform; we verify this via KS uniformity tests at each window as a diagnostic and those results are satisfactory.

4.5 Vine Copula Selection and Fitting

Multivariate dependence among PIT-transformed marginals is modelled using vine copulas (pair-copula constructions), considering Student- t and BB7 bivariate families. Copula selection follows a three-stage procedure:

1. **Goodness-of-Fit:** The Rosenblatt transformation-based Cramér–von Mises (CvM) test (Genest et al., 2009) is applied; a copula is accepted if the resulting p -value exceeds 0.05.
2. **Tail Dependence Consistency:** The empirical lower tail dependence λ_L is estimated from the PIT-transformed data. The model-implied tail dependence λ_{model} is obtained via Monte Carlo simulation. A copula is accepted if: (i) the empirical λ_L lies within the 95% bootstrap confidence interval of the model (based on 1,000 replications of 500 simulations each), and (ii) $|\lambda_{\text{model}} - \lambda_{\text{emp}}| \leq 0.05$.
3. **Rolling Cross-Validation (FZ0-based):** Candidate copulas passing the first two stages are compared using mean FZ0 loss over the last $L = 50$ rolling windows. For initial windows where fewer than L past observations are available, AIC is used as a fallback selection criterion.

Table 4.1 shows that moderate variations in L (e.g., $L = 25$, $L = 50$, $L = 100$) do not materially affect the final mean FZ0 loss of model much (Absolute difference between $L=25$, $L=50$ and $L=50$, $L=100$ has maximum value of around 0.006), which can be seen in Table 4.1, suggesting that the results are not highly sensitive to the specific choice of L . So finally, we considered L as 50 for evaluation.

Table 4.1: Sensitivity of Mean FZ0 Loss to the Choice of Rolling Selection Window L

Horizon	Setting	$L = 25$			$L = 50$			$L = 100$		
		Base	No-KDE	KDE	Base	No-KDE	KDE	Base	No-KDE	KDE
H=1	S=1, 2A	1.1293	1.1307	1.1299	1.1293	1.1318	1.1307	1.1293	1.1314	1.1304
H=1	S=10, 2A	0.9110	0.9327	0.9322	0.9110	0.9334	0.9331	0.9110	0.9367	0.9361
H=1	S=1, 4A	1.0197	1.0170	1.0168	1.0197	1.0170	1.0177	1.0197	1.0144	1.0150
H=1	S=10, 4A	1.0232	1.0161	1.0170	1.0232	1.0226	1.0233	1.0232	1.0280	1.0285
H=10	S=1, 2A	2.2688	2.2668	2.2642	2.2688	2.2670	2.2643	2.2688	2.2670	2.2649
H=10	S=10, 2A	2.3056	2.2761	2.2688	2.3056	2.2758	2.2686	2.3056	2.2767	2.2694
H=10	S=1, 4A	2.2439	2.2577	2.2569	2.2439	2.2587	2.2577	2.2439	2.2602	2.2588
H=10	S=10, 4A	2.2639	2.2708	2.2708	2.2639	2.2715	2.2720	2.2639	2.2735	2.2735

Notes: H = forecast horizon; S = refit step size; A = number of assets.

4.6 VaR and ES Computation

Let h denote the forecast horizon and s the refit step size, with $h \in \{1, 10\}$ and $s \in \{1, 10\}$. VaR and ES at confidence level $\alpha = 5\%$ are computed for an equal-weighted portfolio using Monte Carlo simulation.

One-day horizon ($h = 1$): 10,000 joint residuals are simulated from the selected vine copula and transformed into asset returns using the inverse hybrid CDF along with current conditional mean and volatility. VaR is the α -quantile (5th percentile) of simulated portfolio returns; ES is the average of returns below the VaR threshold.

Ten-day horizon ($h = 10$): Simulation is performed sequentially; at each step, residuals are simulated, transformed into returns, and the GJR-GARCH recursion is used to update conditional mean and variance. The cumulative 10-day return is obtained by summing simulated returns across the horizon, capturing the propagation of volatility dynamics over time.

4.7 Evaluation Framework

4.7.1 Primary Criterion: Mean FZ0 Joint Loss

The primary model comparison criterion is the mean FZ0 joint VaR–ES loss of Fissler and Ziegel (2016):

$$\text{FZ0}(\text{VaR}, \text{ES}; y_t) = \frac{1}{\alpha \text{ES}_t^\alpha} I_t^\alpha(y_t - \text{VaR}_t^\alpha) + \frac{\text{VaR}_t^\alpha}{\text{ES}_t^\alpha} + \log(-\text{ES}_t^\alpha) - 1 \quad (4.21)$$

where $I_t^\alpha = \mathbf{1}\{y_t \leq \text{VaR}_t^\alpha\}$ is the indicator function. Here losses are represented as negative returns, accordingly VaR and ES will be negative. A lower FZ0 loss indicates superior joint calibration of both VaR and ES. Mean FZ0 is the primary criterion. The FZ0 comparison is valid across all eight experimental configurations since it is computed observation-by-observation and does not require non-overlapping windows.

4.7.2 Secondary Criteria: Formal Backtests

Secondary evaluation uses the following tests:

- **Kupiec (1995) likelihood-ratio test:** tests whether the empirical violation rate matches the nominal $\alpha = 5\%$.
- **Christoffersen (1998) independence test:** tests whether VaR violations are serially independent.

- **Dynamic Quantile test** (Engle and Manganelli, 2004) with lags 1 and 4: tests for absence of autocorrelation in the hit sequence.
- **Acerbi–Székely (2014) ES test:** the null hypothesis that the ES model is correctly specified is tested by generating 10,000 null datasets using a global seed of 42.

For $h = 10$, step-size-1 configurations produce 2,462 overlapping observations. The overlap implies that consecutive forecasts share a large portion of underlying data, inducing strong serial dependence in the violation sequence. As a result, the independence assumption underlying the Christoffersen and DQ tests is violated, rendering these tests invalid as formal hypothesis tests; they are therefore reported as diagnostic measures only. The Kupiec test is less sensitive to dependence but may suffer from reduced effective sample size, leading to distorted p -values. The Acerbi–Székely test relies on a Monte Carlo null distribution generated under an i.i.d. assumption; in the presence of overlap-induced dependence, this may result in size distortions, and the test is therefore interpreted with caution. For $h = 10$, step-size-10, and for $h = 1$ in both step sizes there will be only non-overlapping observations, so all tests are valid.

Chapter 5

Empirical Results

This chapter presents the empirical results in four steps. First, models are compared using FZ0 loss as the primary ranking criterion. Second, fallback frequency is examined. Third, formal backtesting results are reviewed as supporting evidence. Finally, overlapping multi-day results are discussed.

5.1 Model Comparison Based on FZ0

Table 5.1 reports results across all eight configurations. Differences in mean FZ0 between models are small (maximum approximately 0.037).

Table 5.1: Full Backtesting Results Across All Eight Experimental Configurations

Horizon	Setting	Model	Viol. Rate	Kupiec p	Christoffersen p	DQ p (L1)	DQ p (L4)	Acerbi p	FZ0 Mean
H=1	S=1, 2A	Baseline	4.29%	0.097	0.784	0.263	0.667	0.511	1.1293
		No-KDE	5.02%	0.967	0.924	0.994	0.913	0.654	1.1318
		KDE	4.94%	0.886	0.991	0.990	0.955	0.620	1.1307
H=1	S=10, 2A	Baseline	3.63%	0.299	0.409	0.544	0.149	0.217	0.9110
		No-KDE	4.84%	0.907	0.268	0.727	0.389	0.366	0.9334
		KDE	4.84%	0.907	0.268	0.727	0.389	0.365	0.9331
H=1	S=1, 4A	Baseline	4.45%	0.203	0.348	0.296	0.263	0.622	1.0197
		No-KDE	4.82%	0.673	0.345	0.569	0.428	0.534	1.0170
		KDE	4.78%	0.606	0.323	0.521	0.404	0.514	1.0177
H=1	S=10, 4A	Baseline	3.23%	0.172	0.464	0.408	0.798	0.376	1.0232
		No-KDE	3.63%	0.299	0.409	0.544	0.836	0.382	1.0226
		KDE	3.63%	0.299	0.409	0.544	0.836	0.381	1.0233
H=10 [†]	S=1, 2A	Baseline	3.86%	0.007 [‡]	0.000*	0.000*	0.000*	0.147	2.2688
		No-KDE	4.39%	0.154	0.000*	0.000*	0.000*	0.438	2.2670
		KDE	4.31%	0.106	0.000*	0.000*	0.000*	0.397	2.2643
H=10	S=10, 2A	Baseline	3.64%	0.305	0.320	0.363	0.616	0.405	2.3056
		No-KDE	4.05%	0.479	0.406	0.542	0.644	0.471	2.2758
		KDE	4.05%	0.479	0.406	0.542	0.644	0.464	2.2686
H=10 [†]	S=1, 4A	Baseline	5.00%	0.993	0.000*	0.000*	0.000*	0.910	2.2439
		No-KDE	5.52%	0.240	0.000*	0.000*	0.000*	0.977	2.2587
		KDE	5.40%	0.366	0.000*	0.000*	0.000*	0.966	2.2577
H=10	S=10, 4A	Baseline	4.45%	0.688	0.451	0.587	0.693	0.642	2.2639
		No-KDE	5.67%	0.637	0.760	0.928	0.648	0.777	2.2715
		KDE	5.67%	0.637	0.760	0.928	0.648	0.776	2.2720

Notes: H = forecast horizon; S = refit step size; A = number of assets. [†] indicates overlapping forecasts, for which independence-based tests are reported for diagnostic purposes only. * denotes structurally degenerate values caused by serial dependence induced by overlapping observations. [‡] denotes rejection of the Kupiec test at the 5% significance level.

No single model uniformly dominates across all configurations; instead, model performance varies across horizons and portfolio sizes.

At the one-day horizon ($h = 1$), the Gaussian baseline performs best for the two-asset portfolio under both step sizes, while the no-KDE variant achieves the lowest FZ0 for the four-asset portfolio.

At the ten-day horizon ($h = 10$), the KDE variant performs best for the two-asset portfolio, whereas the baseline model performs best for the four-asset portfolio under both step sizes.

These results indicate that more complex tail modelling does not consistently outperform the simpler baseline. The ranking varies across configurations, suggesting that model performance is context-dependent rather than driven by a uniform advantage of any one specification.

5.2 Fallback Frequency and EVT Stability

Table 5.2 summarises EVT threshold selection outcomes across assets and configurations. Fallback occurs in a non-trivial fraction of windows, ranging from 0% to 16.35%, confirming that EVT threshold failure is not negligible and handling fallback is important.

Table 5.2: EVT Threshold Selection Outcomes by Asset and Configuration

Asset	Setting	Tail	Total Windows	Invalid	Sparse	Fallback Rate
INFY.NS	S=1	Lower	2471	0	0	0.00%
INFY.NS	S=1	Upper	2471	282	122	16.35%
SBIN.NS	S=1	Lower	2471	26	322	14.08%
SBIN.NS	S=1	Upper	2471	27	13	1.62%
INFY.NS	S=10	Lower	248	0	0	0.00%
INFY.NS	S=10	Upper	248	26	14	16.13%
SBIN.NS	S=10	Lower	248	3	34	14.92%
SBIN.NS	S=10	Upper	248	4	1	2.02%
TATAMOTORS.NS	S=1	Lower	2471	61	157	8.82%
TATAMOTORS.NS	S=1	Upper	2471	3	14	0.69%
HINDUNILVR.NS	S=1	Lower	2471	25	53	3.16%
HINDUNILVR.NS	S=1	Upper	2471	120	63	7.41%
HINDUNILVR.NS	S=10	Lower	248	3	5	3.23%
HINDUNILVR.NS	S=10	Upper	248	14	4	7.26%
TATAMOTORS.NS	S=10	Lower	248	7	14	8.47%
TATAMOTORS.NS	S=10	Upper	248	0	1	0.40%

Notes: Fallback Rate = (invalid + sparse) / total $\times$ 100.

There is substantial heterogeneity across assets and between upper and lower tails. In some cases, fallback is driven primarily by invalid thresholds, indicating failure of the GPD fit, while in others it is driven by sparse exceedances, indicating insufficient tail data. These patterns imply that EVT applicability varies across assets, tails, and time, rather than being uniformly reliable.

With-KDE and no-KDE variants differ only in windows where EVT cannot be reliably applied. The with-KDE model replaces the affected tail using a nonparametric KDE approach, while the no-KDE model relies on the parametric distribution implied by the GJR-GARCH specification. As a result, differences between these two models arise exclusively in fallback windows. Although fallback occurs in only a subset of windows, its impact is not negligible. In some configurations, the KDE-based approach achieves lower FZ0, while in others the no-KDE approach performs better. This indicates that the way fallback is handled can influence model performance, even if overall differences remain small.

5.3 Backtesting Results

At the one-day horizon, all models pass the standard backtests. Differences in performance are therefore driven primarily by efficiency as captured by FZ0, rather than by violations of statistical assumptions.

At the ten-day horizon under non-overlapping evaluation ($h = 10, s = 10$), all models again satisfy the backtesting criteria, confirming broadly well-calibrated VaR and ES forecasts. An exception arises in the overlapping setting ($h = 10, s = 1$) where the baseline model fails the Kupiec test in the two-asset configuration; however, this must be interpreted with caution due to overlap-induced dependence.

5.4 Overlapping Multi-Day Results

The overlapping setting ($h = 10, s = 1$) produces a larger number of forecast observations, but these are strongly dependent due to shared underlying return paths. As a result, independence-based backtests such as the Christoffersen and Dynamic Quantile (DQ) tests yield structurally degenerate outcomes, while the Kupiec and Acerbi–Székely tests may also experience size distortions due to the reduced effective sample size and violation of underlying independence assumptions. Consequently, FZ0 remains the primary evaluation metric.

Chapter 6

Discussion

The empirical results highlight that EVT threshold failure is not a marginal issue in rolling risk estimation, but a recurring feature that must be explicitly addressed. Fallback is triggered in up to 16.35% of rolling windows, with substantial variation across assets and between upper and lower tails. This indicates that EVT-based tail modelling cannot be assumed to be uniformly applicable over time, and that any practical implementation must incorporate a mechanism to handle periods in which threshold selection is unreliable.

The role of fallback becomes particularly relevant when comparing the with-KDE and no-KDE variants within the EVT-based framework. These two models are identical in windows where EVT threshold selection succeeds; any differences between them arise only in fallback windows. The results show that neither fallback approach consistently dominates: in some configurations the KDE-based fallback achieves lower FZ0, while in others the no-KDE approach performs better. This indicates that the way fallback is implemented can influence model performance, even though its overall effect remains limited due to the relatively small proportion of fallback occurrences.

Differences in FZ0 across models are consistently small, and no single model provides uniformly superior performance. While EVT-based models offer improvements in certain configurations, the Gaussian baseline remains competitive throughout. This suggests that performance depends on the interaction between horizon length and portfolio dimension, rather than on a single dominant modelling approach.

The overlapping multi-day setting further illustrates the importance of careful evaluation. The dependence induced by overlapping observations limits the interpretability of standard backtesting procedures, particularly those based on independence assumptions. In this context, FZ0 provides a more reliable basis for comparison.

Chapter 7

Managerial Implications

The results highlight several practical considerations for the design and implementation of risk systems.

Fallback-aware risk systems: Risk systems based on EVT should incorporate an explicit fallback policy to ensure continuity of VaR and ES forecasts when threshold selection fails. The choice of fallback method can affect results in these periods, even if overall performance differences remain small.

Monitoring fallback as a model-health indicator: Since fallback is triggered in a measurable subset of windows and varies across assets and tails, tracking fallback frequency provides useful diagnostic information. High or persistent fallback rates may indicate instability in tail behaviour or limitations of the EVT specification, and can serve as a model-health indicator in operational risk systems.

Overlap-aware evaluation for multi-day risk: For multi-day horizons, overlapping observations introduce strong dependence across forecasts. Standard backtesting procedures based on independence assumptions can produce misleading results in this setting. Risk systems should adopt overlap-aware evaluation protocols, distinguishing between diagnostic and formally valid tests, and relying more heavily on metrics such as FZ0 that do not require independence.

Backtests versus scoring-based evaluation: The results show that passing standard backtests does not necessarily imply superior model performance. All models achieve broadly acceptable calibration, yet differences remain in FZ0. This highlights that regulatory backtesting and scoring-rule-based evaluation serve different purposes: the former ensures adequacy, while the latter provides a more sensitive ranking of competing models.

Chapter 8

Conclusions and Future Scope

8.1 Conclusions

This report develops and evaluates a multi-horizon portfolio risk estimation framework combining AR(1)–GJR-GARCH filtering, automated EVT threshold selection, and vine copula dependence modelling in a rolling-window setting applied to Indian equity markets.

The empirical results establish that EVT threshold failure is operationally relevant in practice. Across assets and tails, valid thresholds cannot be consistently identified in all rolling windows, with fallback triggered in up to 16.35% of cases. This confirms that EVT-based tail modelling cannot be assumed to apply uniformly over time, and that any practical implementation must explicitly account for threshold instability.

The comparison of models shows that no single specification uniformly dominates across all horizons and portfolio configurations. Differences in FZ0 are consistently small, and relative performance varies across settings. While EVT-based models provide improvements in some cases, the Gaussian baseline remains competitive throughout.

Within the EVT-based framework, the choice of fallback policy can influence results in the subset of windows where threshold selection is not reliable. The with-KDE and no-KDE approaches perform differently across configurations, with neither approach consistently outperforming the other. This indicates that how the fallback is designed is an important modelling decision.

The results also emphasise the importance of appropriate evaluation protocols. For multi-day horizons, overlapping observations introduce dependence that invalidates standard independence-based backtests such as Christoffersen and Dynamic Quantile. In such settings, evaluation should rely more heavily on measures such as FZ0 that remain valid under dependence.

8.2 Future Scope

Future work may extend this analysis in several directions:

1. **Larger asset universes:** Extending the framework to portfolios containing a larger number of assets would provide further insight into the scalability of vine copula estimation and fallback-aware risk modelling.

2. **Stress-period analysis:** Dedicated analysis of major stress periods like global financial crisis, would provide additional evidence regarding the robustness of fallback-aware EVT modelling under extreme market conditions.
3. **Alternative tail modelling approaches:** Comparing the proposed EVT-based framework with alternative risk forecasting approaches, including machine-learning-based tail estimation methods and neural-network architectures, could provide additional insight into modelling trade-offs between flexibility, interpretability, and statistical robustness.
4. **Real-time implementation:** Developing a production-oriented implementation capable of monitoring fallback frequency and EVT threshold reliability in real time could further improve the practical applicability of the framework for institutional risk management.

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Appendix A

Results and Code

Complete outputs,report and implementation code are available at: <https://github.com/NARAYANASETTIDEVENDRA1/Robust-Multi-Asset-VaR-and-Expected-Shortfall-Forecasting->